Yongjin Choi

Assistant Professor
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EDUCATION

Ph.D. in Civil Engineering 2024
University of Texas at Austin, Department of Civil, Architectural and Environmental Engineering, USA
- Specialty area: Geotechnical engineering
- Dissertation: *Forward and Inverse Modeling of Granular Flows Using Differentiable Graph Neural Network Simulator*

M.S. in Civil Engineering 2019
Pusan National University, Department of Civil and Environmental Engineering, South Korea
- Specialty area: Geotechnical engineering
- Thesis: *Stiffness Evaluation of Open-Graded Aggregate Base Corresponding to Compaction Level Based on Plate Load Tests*

B.S. in Civil Engineering 2017
Pusan National University, Department of Civil and Environmental Engineering, South Korea

APPOINTMENTS

Assistant Professor (Incoming) Sep 2026
University of Massachusetts Amherst, Department of Civil and Environmental Engineering, USA

Postdoctoral Fellow 2025–2026
Korea Advanced Institute of Science and Technology (KAIST), InnoCORE PRISM-AI Center, Department of Mechanical Engineering, South Korea

Affiliated Research Fellow 2025–2026
Georgia Institute of Technology, School of Civil and Environmental Engineering, USA

Postdoctoral Fellow 2024–2025
Georgia Institute of Technology, School of Civil and Environmental Engineering, USA

Graduate Research Assistant 2021–2024
University of Texas at Austin, Department of Civil, Architectural and Environmental Engineering, USA

Graduate Research Assistant 2017–2020
Pusan National University, Department of Civil and Environmental Engineering, South Korea

PUBLICATIONS

† Equal contribution * Corresponding author

Peer-Reviewed Journal Articles

[14] **Choi, Y.**, Lee, S.* (2026). Accelerating stochastic simulation of post-failure landslide runout using a random graph neural network-based simulator. *Computers and Geotechnics*, 195, 108104. DOI

[13] **Choi, Y.**, Liu, C., Macedo, J.* (2026). Operator learning for consolidation: An architectural comparison for DeepONet variants. *Computers and Geotechnics*, 194, 108017. DOI

[12] **Choi, Y.**, Macedo, J.*, Liu, C. (2026). Differentiable graph neural network simulator for forward and inverse modeling of multi-layered slope system with multiple material properties. *Soils and Foundations*, 66(3), 101773. DOI

[11] Ryou, J.-E., Gang, S., **Choi, Y.**, Dai, S. C., Jung, J.* (2026). The interplay between pore and particle characteristics in sands based on computed tomographic image analyses. *Powder Technology*, 473, 122150. DOI

[10] Bae, B., **Choi, Y.***, Jung, H., Ahn, J.* (2025). Tunnel lining segmentation from ground-penetrating radar images using advanced single-and two-stage object detection and segmentation models. *Computer-Aided Civil and Infrastructure Engineering*, 40(22), 3580–3592. DOI

[9] Kim, T., Shahriar, A. R., Lee, W.-D.*, **Choi, Y.**, Kwon, S., Gabr, M. A. (2025). Field data-based prediction of local scour depth around bridge piers using interpretable machine learning. *Transportation Geotechnics*, 52, 101567. DOI

[8] **Choi, Y.**, Bae, B., Han, T. H., Ahn, J.* (2024). Application of Mask R-CNN and YOLOv8 algorithms for concrete crack detection. *IEEE Access*, 12, 165314–165321. DOI

[7] **Choi, Y.***, Kumar, K. (2024). Graph neural network-based surrogate model for granular flows. *Computers and Geotechnics*, 166, 106015. DOI CODE

[6] **Choi, Y.***, Kumar, K. (2024). Inverse analysis of granular flows using differentiable graph neural network simulator. *Computers and Geotechnics*, 171, 106374. DOI CODE

[5] **Choi, Y.**, Nguyen, H.-T., Han, T. H., Choi, Y., Ahn, J.* (2024). Sequence deep learning for seismic ground response modeling: 1D-CNN, LSTM, and transformer approach. *Applied Sciences*, 14(15), 6658. DOI CODE

[4] Lee, Y., Lee, D., Lee, H., Choe, H.-S., Kim, J.-H., **Choi, Y.***, Ahn, J.* (2024). Removal Efficiency of Bottom Ash and Sand Mixtures as Filter Layers for Fine Particulate Matter. *Materials*, 17(11), 2749. DOI

[3] Lee, Y., **Choi, Y.**, Ahn, D., Ahn, J.* (2021). Prediction models based on regression and artificial neural network for moduli of layers constituted by open-graded aggregates. *Materials*, 14(5), 1199. DOI

[2] **Choi, Y.**, Ahn, D., Lee, Y., Ahn, J.* (2020). Compaction quality monitoring of open-graded aggregates by light weight deflectometer and soil stiffness gauge. *Sustainability*, 12(6), 2521. DOI

[1] **Choi, Y.**, Ahn, D., Nguyen, T. H., Ahn, J.* (2018). Assessment of field compaction of aggregate base materials for permeable pavements based on plate load tests. *Sustainability*, 10(10), 3817. DOI

Manuscripts Under Review

[7] **Choi, Y.**, Macedo, J.* (2026). Differentiable Graph Neural Network Simulator for the Back-Analysis of Post-Liquefaction Residual Strength from Flow Failure Runout. Under review in *Canadian Geotechnical Journal*. PREPRINT

[6] **Choi, Y.**[†], Moon, H.[†], Ryu, S.[*] (2026). Critical evaluation of PINN for FWD inverse analysis and differentiable FEM as an alternative. Under review. PREPRINT CODE

[5] **Choi, Y.**, Ok, S., Lee, J., Ryu, S.[*], Choo, J.[*] (2026). An agentic-AI calibrate–diagnose–suggest workflow for parameter identification in soil constitutive models. Under review in *Computers and Geotechnics*.

[4] Kim, T.[†], **Choi, Y.**[†], Kwon, S., Kim, J.[*] (2026). An interpretable DeepONet surrogate for real-time prediction of three-dimensional urban surface temperature fields. Under review in *Sustainable Cities and Society*.

[3] Moon, H.[†], **Choi, Y.**[†], Ryu, S.[*] (2026). A Hybrid GNN-FEM Framework for Phase-Field Fracture Simulation. Physics-Preserving Hybridization for Generalizable Surrogate Modeling. Under review in *Engineering Fracture Mechanics*. PREPRINT

[2] Nguyen, H.-T., **Choi, Y.**, Kang, S., Ahn, J.[*] (2026). A low-cost Raspberry Pi data acquisition system for lightweight deflectometers with synchronized raw-signal access. Under review in *IEEE Sensors*.

[1] Ryou, J.-E., Gang, S., **Choi, Y.**, Lee, E. H., Jung, J.[*] (2026). Microstructure-informed prediction of hydraulic conductivity using pore network modeling and interpretable surrogate learning. Under review in *Journal of Hydrology*.

Conference Papers

[4] **Choi, Y.**, Kumar, K. (2024). Three-dimensional granular flow simulation using graph neural network-based learned simulator. In *Geo-Congress 2024*, pp. 335–344. DOI

[3] **Choi, Y.**, Kumar, K. (2023). A machine learning approach to predicting pore pressure response in liquefiable sands under cyclic loading. In *Geo-Congress 2023*, pp. 202–210. DOI CODE

[2] Kumar, K., **Choi, Y.** (2023). Accelerating particle and fluid simulations with differentiable graph networks for solving forward and inverse problems. In *Proceedings of the SC'23 Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis*, pp. 60–65. DOI

[1] **Choi, Y.**, Lee, S., Oh, J., Ahn, J., Jung, J. (2018). Coefficient of subgrade reaction for the permeable block and base system at Korea GI and LID center. In *Solving Pavement and Construction Materials Problems with Innovative and Cutting-edge Technologies: Proceedings of the 5th GeoChina International Conference 2018*, pp. 69–75. DOI

Peer-Reviewed Korean Journal Articles

[6] Bae, B., **Choi, Y.**, Yun, K., Ahn, J. (2024). Application of Mask R-CNN Algorithm to Detect Cracks in Concrete Structure. *Journal of Korean Geotechnical Society*, 40(3), 33–39. DOI

[5] **Choi, Y.**, Son, S. (2024). Sensitivity analysis of factors influencing the settlement of buried pipes installed beneath roads using explainable machine learning techniques. *Journal of the Korean Asphalt Institute*, 14(2), 269–278. DOI

[4] Jung, J., **Choi, Y.**, Ahn, J. (2021). Evaluation of p-y Curves for Group Piles in Cohesive Soil Based on 3D Numerical Analysis. *Journal of the Korean Society of Hazard Mitigation*, 21(6), 193–200. DOI

[3] **Choi, Y.**, Ahn, J.[*] (2020). Evaluation of p-y curves and p-multiplier of pile groups corresponding to sand properties change based on 3D numerical analysis. *Journal of the Korean Society of Hazard Mitigation*, 20(4), 207–217. DOI CODE

[2] **Choi, Y.**, Nguyen, T. H., Ahn, J.[*] (2019). Resilient Modulus and Permanent Deformation of 40-mm Open-Graded Aggregates based on Repeated-Load Triaxial Tests. *Journal of the Korean Society of Hazard Mitigation*, 19(7), 367–374. DOI

[1] **Choi, Y.**, Oh, J.-H., Han, S.-I., Ahn, J., Shin, H.-S.* (2017). Elastic modulus and layer coefficient of permeable block pavements based on plate load tests. *Journal of the Korean Geotechnical Society*, 33(12), 75–80. DOI

CONFERENCE PRESENTATIONS

Underline: presenter

[7] Alam, M. M., **Choi, Y.**, Yousaf, A., Seong, W. K., Ryu, S., Ruoff, R. S. (2026). Geometry of a non-centrosymmetric 'truncated octahedral' diamond particle. *Korean Institute of Metals and Materials Spring Conference 2026*, Jeju, South Korea (Poster).

[6] **Choi, Y.**, Lee, S., Ryu, S. (2026). Accelerating stochastic simulation of post-failure landslide runout using a random graph neural network-based simulator. *17th World Congress on Computational Mechanics and 10th European Congress on Computational Methods in Applied Sciences and Engineering (WCCM-ECCOMAS 2026)*, Munich, Germany (Oral).

[5] **Choi, Y.**, Cornejo, R., Macedo, J., Kumar, K. (2024). AI/MPM-based assessment of dam break failure modes and runout. *TAILENG Annual Meeting*, Denver, USA (Oral).

[4] **Choi, Y.**, Kumar, K. (2023). Slope inverse analysis using accurate and generalizable data-driven surrogate granular flow simulator. *17th U.S. National Congress on Computational Mechanics*, Albuquerque, USA (Oral).

[3] Kumar, K., Mehta, C., **Choi, Y.** (2023). Graph network simulator and differentiable material point method. *Advances in Computational Mechanics 2023*, Austin, USA (Oral).

[2] **Choi, Y.**, Ahn, D., Nguyen, H. T., Ahn, J. (2018). Measurement of elastic modulus of 40 mm crushed stone aggregate with field compaction effort. *Korean Society of Railway Spring Conference 2018*, Jeju, South Korea (Oral).

[1] **Choi, Y.**, Jalmasco, M. M., Ahn, J. (2017). Permeability degradation of permeable pavement material due to clogging. *JpGU-AGU Joint Meeting 2017*, Chiba, Japan (Oral).

HONORS & AWARDS

- Bright Spark Lecture Award, International Society for Soil Mechanics and Geotechnical Engineering (ISSMGE), presented at the 4th International Workshop on the Future of Machine Learning in Geotechnics (4FOMLIG) — 2026
- Community of Practice (CoP) Program Fellowship, Korea Institute of Human Resources Development in Science and Technology (KIRD) — 2026
- Young Scholar Award, Korean Geotechnical Society — 2023
- Second Best Poster Presentation Award, Graduate Student Symposium of Civil, Architectural, and Environmental Engineering, University of Texas at Austin — 2022
- Frontera Compute Resource Award (under the supervision of Dr. Krishna Kumar), Texas Advanced Computing Center (TACC) — 2021
- Outstanding Graduate Student Research Paper Award, Brain Korea 21 Plus — 2019
- Fellowship, "Earthquake Disaster Prevention Human Resource Development" Program, Korean Ministry of the Interior and Safety — 2017–2018
- Scholarship for Academic Excellence, Korea Student Aid Foundation — 2015–2016

TEACHING

Guest Lecturer Fall 2025

Deep Learning and Its Application to Geotechnical Systems, Pusan National University, South Korea

- Delivered a two-hour lecture on physics-informed neural networks and graph neural networks for geotechnical modeling for graduate students.

Assistant Instructor Spring 2025

Numerical Modeling and Data Science for Geotechnics (CEE 8813-H), Georgia Institute of Technology, USA

- Designed and delivered lectures and course materials on material point methods, finite element methods, and regression and classification machine learning models for geotechnical engineers, which are publicly available through open-source websites.

Teaching Assistant Spring 2023

Introduction to Computer Methods (CE 311K), University of Texas at Austin, USA

- Hosted weekly laboratory sessions for hands-on Python coding activities for engineering problems, held regular office hours, and managed grading tasks.

Teaching Assistant Spring 2021

Probability and Statistics for Civil Engineers (CE 311S), University of Texas at Austin, USA

- Designed and hosted weekly laboratory sessions with hands-on problem-solving activities, held regular office hours, and managed grading tasks.

INVITED TALKS

- Department of Civil, Urban and Environmental Engineering, Seoul National University, South Korea — Sep 2026
- RIT Seminar, Department of Civil and Environmental Engineering, Yonsei University, South Korea — May 2026
- Geosystem Lab, Department of Civil, Environmental and Architectural Engineering, Korea University, South Korea — Mar 2026
- School of Civil Engineering, Chungbuk National University, South Korea — Jan 2026
- Korea Institute of Geoscience and Mineral Resources (KIGAM), South Korea — Dec 2025
- Department of Civil, Urban, Earth, and Environmental Engineering, Ulsan National Institute of Science and Technology (UNIST), South Korea — Dec 2025
- Computational Structural Mechanics and Design Lab, Department of Mechanical Engineering, Ulsan National Institute of Science and Technology (UNIST), South Korea — Nov 2025
- GeoKAIST Seminar, Department of Civil and Environmental Engineering, Korea Advanced Institute of Science and Technology (KAIST), South Korea — Oct 2025

PROFESSIONAL SERVICE

Peer Review

Journal Reviewer 2021–Present

Journal of Computing in Civil Engineering; Journal of Rock Mechanics and Geotechnical Engineering; Earthquake Spectra; Journal of Traffic and Transportation Engineering; Chemical Engineering Journal; Journal of Mechanical Science and Technology; International Journal of Geo-Engineering

Committees and Advisory Boards

Advisory Board Member, Revolutionizing Engineering Departments (RED) Program 2025

Georgia Institute of Technology, USA

- Engaged in ongoing educational innovation to incorporate computational thinking and artificial intelligence into the Civil and Environmental Engineering program.

SOFTWARE DEVELOPMENT

Learn-MPM-2D WEBSITE 2024

- Developed an open-source learning package for the 2-D material point method, guiding learners from core theory to hands-on practice in a beginner-friendly interactive Python notebook environment.

Cbgeopy: Material Point Method Model Generator DOCS 2024

- Developed a material point method model generator that supports various geometries from simple columns to multi-layered 3-D topography, which can be integrated with CB-Geo MPM.

GNS: A Generalizable Graph Neural Network-Based Simulator for Particulate and Fluid Modeling CODE 2023

- Contributed as a core developer to implement mesh graph networks for fluid modeling, multi-material features, and code maintenance.

PROFESSIONAL LICENSES & CERTIFICATIONS

- Advanced Teaching Certificate, Center for Teaching and Learning, University of Texas at Austin, USA 2024
- Engineer in Training (EIT), Civil Engineering, NCEES, USA 2023
- License of Engineer (FE), Civil Engineering, Human Resources Development Service of Korea 2016